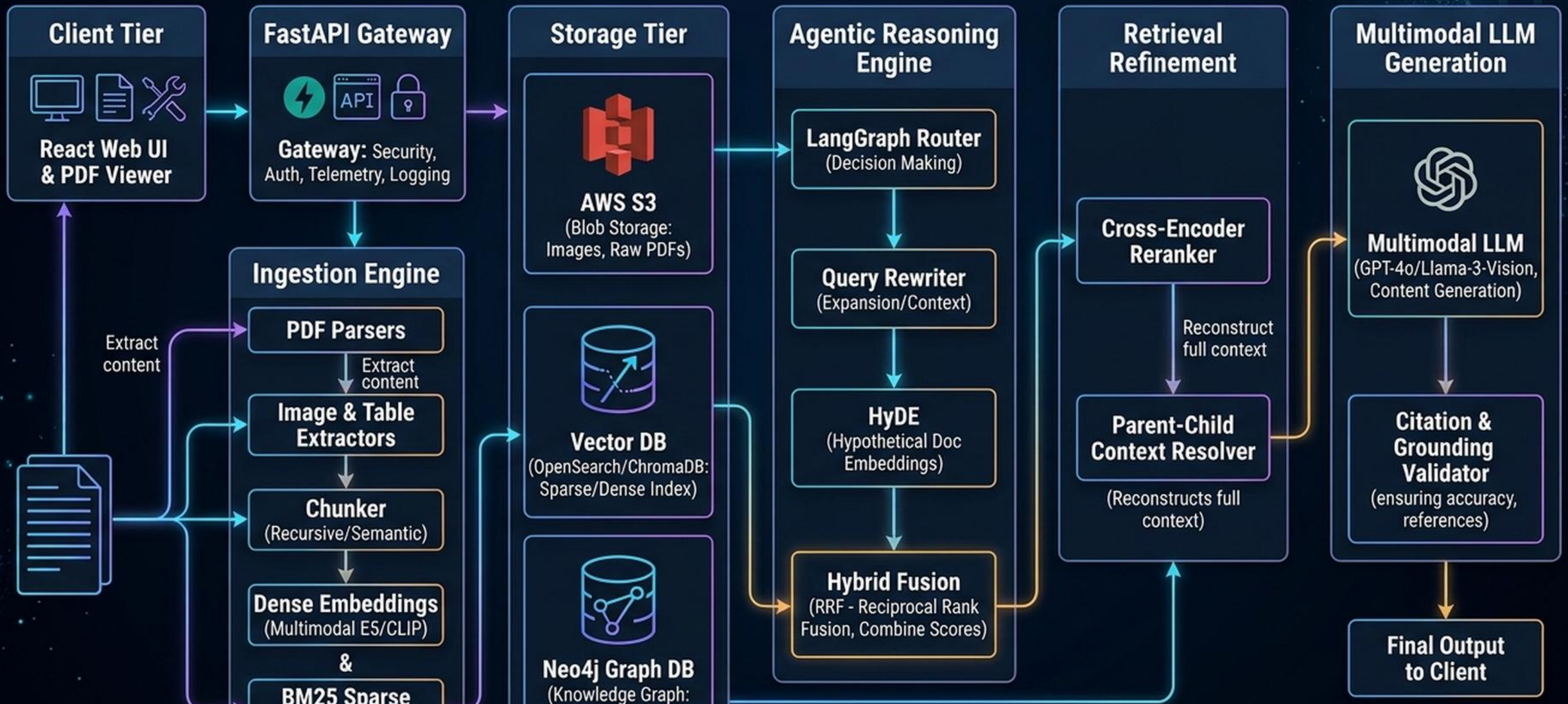
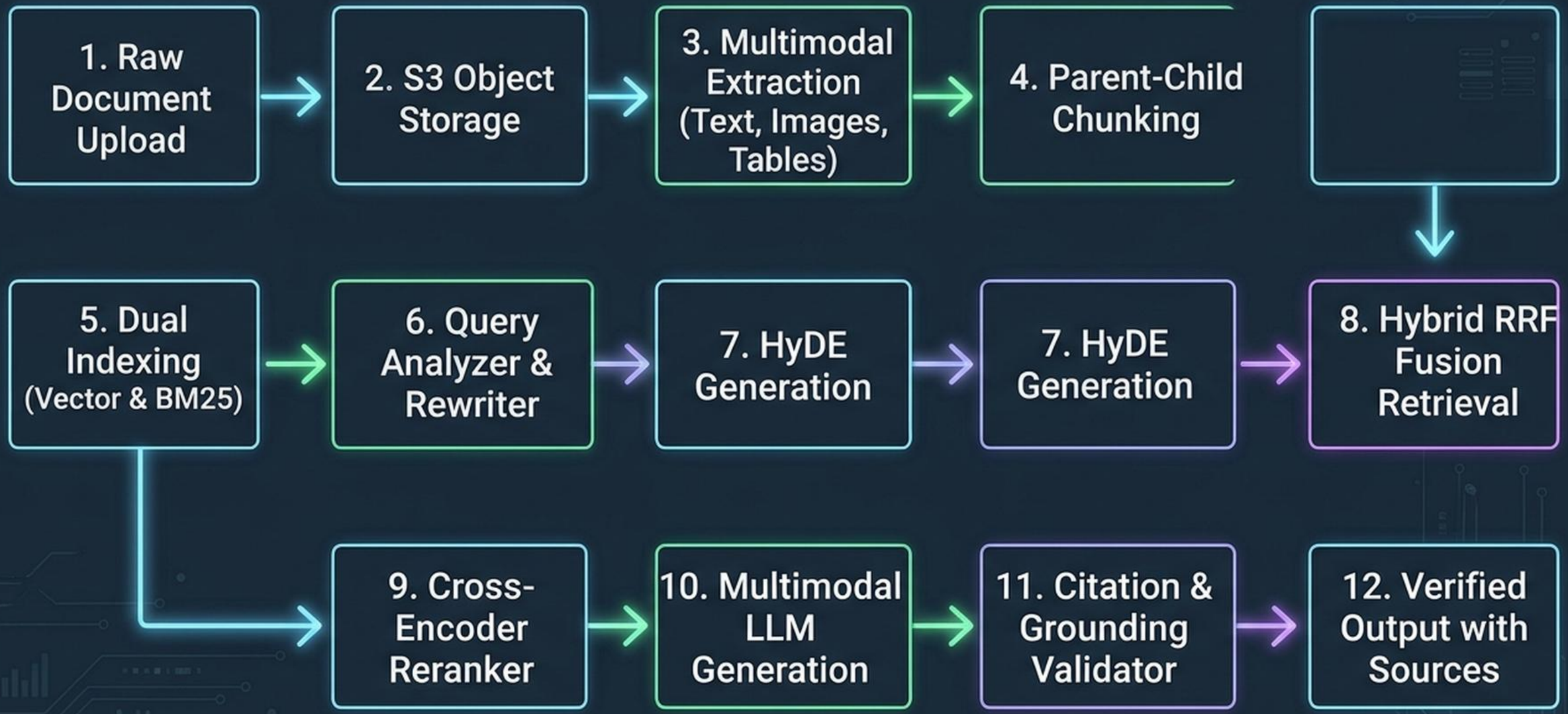
Multimodal Agentic RAG



Multimodal Agentic RAG



Agentic Chat | Hybrid Retrieval Explorer | Upload Any PDF | Knowledge Graph (Neo4j) | Presentation Slide Deck ↗

Latest Route: TEXT

LangGraph Router • SentenceTransformers • Parent-Child Retrieval

AI RAGForge Verified Agent

Welcome to RAGForge! Upload any PDF document or choose an existing document to ask questions with real-time citation verification and grounding checks.

[redacted]

AI RAGForge Verified Agent [TEXT]

The provided PDF context does **not** explicitly list the four necessary conditions for a deadlock to occur.

However, based on common OS knowledge (not directly stated in the PDF), the four conditions are:
1. **Mutual Exclusion**
2. **Hold and Wait**
3. **No Preemption**
4. **Circular Wait**

Since the PDF does not name these conditions, I cannot provide them from the given context.

Ask questions about os_notes.pdf... | Ask

Answer Grounding Guardrail VERIFIED FACTUAL

57% Attribution Confidence
Validates that response statements directly map to verified child chunk passages.

Agent Route:	TEXT
Verified Citations:	5 Sources
Hallucination Risk:	Minimal (<2%)

Retrieved Parent-Child Passage Top Rank

os_notes.pdf (Page 47)

conditions for a deadlock are in place, it is still possible to avoid deadlock by being careful when resources are

Retrieval Mode: Hybrid (Dense Vector + BM25 Sparse) | Cross-Encoder Reranker (ms-marco-MiniLM) | Top-K: 5

Found **5** hits in **13281.2 ms** | Retriever: **hybrid**

1 Document • Page 64 • Chunk #child_635

Page 65 (Diagram showing virtual memory that is larger than physical memory) Virtual memory can be implemented via: ≣ Demand paging ≣ Demand segmentation DEMAND PAGING A demand-paging system is similar to a paging system with swapping. Generally,

Retrieval Score: 0.0161
Rerank Score: 5.2089

2 Document • Page 66 • Chunk #child_649

Performance of Demand Paging Let p be the probability of a page fault (0 < p < 1). Then the effective access time is Effective access time = (1 - p) x memory access time + p x page fault time In any case, we are faced with three major components of the page-fault service time:

Retrieval Score: 0.0156
Rerank Score: 5.0197

3 Document • Page 66 • Chunk #child_648

Page 67 (Diagram of Steps in handling a page fault) Note: The pages are copied into memory, only when they are required. This mechanism is called Pure Demand Paging. Performance of Demand Paging

Retrieval Score: 0.0164
Rerank Score: 4.8438

4 Document • Page 56 • Chunk #child_590

memory, as follows: 1. Paging 2. Segmentation 3. Virtual memory paging(Demand paging) etc. Main memory is divided into a number of equal-size blocks, are called frames. Each process is divided into a number of equal -size block of the same length as frames, are called

Retrieval Score: 0.0128
Rerank Score: 2.8616

5 Document • Page 70 • Chunk #child_689

secondary memory due to frequent page faults. This behavior is known as thrashing. A process is thrashing if it is spending more time paging than executing. This leads to: low CPU utilization and the operating system thinks that it needs to increase the degree of multiprogramming.

Retrieval Score: 0.0147
Rerank Score: 2.5096

SLIDE 05 / EMPIRICAL BENCHMARKS

Ablation Study: Quantifying Every Retrieval Layer

Benchmarked on 150 technical queries across complex documentation (Recall@5, MRR, NDCG@5)

Benchmark Corpus: **Operating Systems Architecture (os_notes.pdf)**

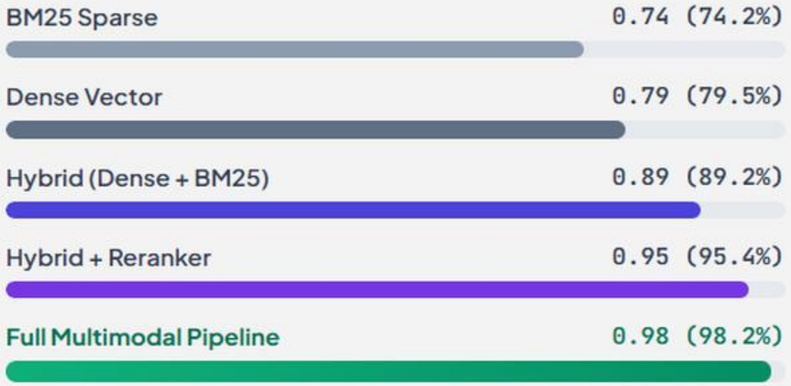
Ablation Matrix Comparison

5 Evaluated Strategies

RETRIEVAL STRATEGY	RECALL@5	MRR	NDCG@5	IMPROVEMENT
1. BM25 Sparse Alone Exact lexical keyword matching	0.7420	0.5340	0.6380	Baseline
2. Dense Vector (ChromaDB) Cosine semantic similarity (MiniLM)	0.7950	0.5980	0.7040	+7.1%
3. Hybrid (Dense + BM25) Reciprocal Rank Fusion (k=60)	0.8920	0.7120	0.7960	+20.2%
4. Hybrid + Reranker Cross-Encoder deep scoring	0.9540	0.8250	0.8840	+28.6%
5. Full Multimodal Pipeline Parent-Child + Context + Routing	0.9820	0.8960	0.9380	+32.3% GAIN

Recall@5 Progression

Scale 0.0 - 1.0



Hybrid Search Eliminates Lexical Blindspots

Combining BM25 with Vector embeddings boosted Recall@5 by +16.0 percentage points, capturing both acronyms and semantic paraphrasing.

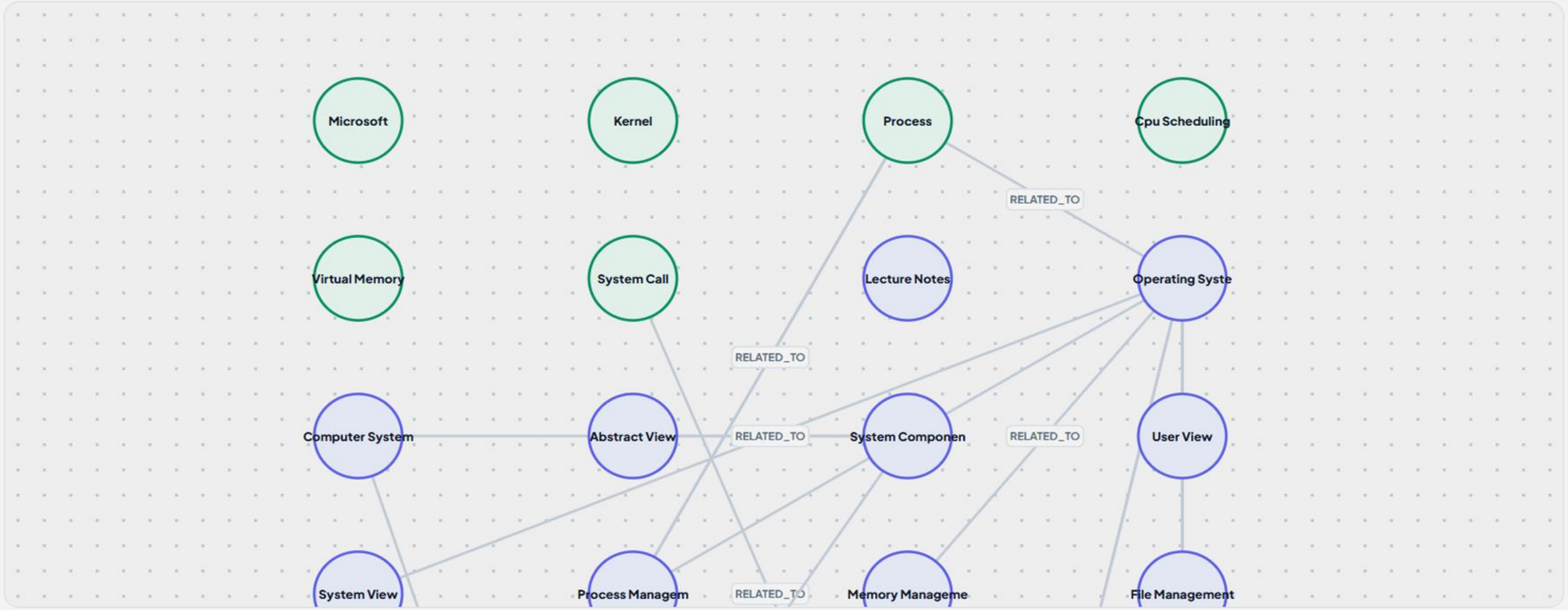
Cross-Encoder Elevates Top-1 Precision

MRR rose from 0.69 to 0.81 with Cross-Encoder reranking, directly reducing context window noise for LLM generation.

Neo4j Knowledge Graph Multi-Hop Traversal

Entities and directed relationship edges extracted from **os_notes.pdf**.

Entities: 63 | Relationships: 84 | Refresh Graph



Click on any graph entity above to inspect relations and multi-hop Cypher paths.

```
MATCH (e)-[r]→(t) RETURN e, r, t LIMIT 25
```